# Linea Mixed Upgrade Audit Report

Prepared by Cyfrin

Version 2.0


**Lead Auditors**

Dacian

Stalin


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# 1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

# 2 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the solidity implementation of the contracts.

# 3 Risk Classification

| | **Impact: High** | **Impact: Medium** | **Impact: Low** |
|---|---|---|---|
| **Likelihood: High** | Critical | High | Medium |
| **Likelihood: Medium** | High | Medium | Low |
| **Likelihood: Low** | Medium | Low | Low |

# 4 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence which aims to provide the "Ethereum experience" at scale and has been previously audited by Cyfrin.

# 5 Audit Scope

This was an incremental audit focused on the following mix of changes:

- dynamic chain variables
- contracts folder restructuring
- data modularization
- transient storage

The audit scope was limited to:

```
1) changes merged from deleted branch `feat/996-dynamic-chain-variables`

// changes to `finalizeBlocks`, `_computePublicInput`
contracts/src/rollup/LineaRollupBase.sol
// move of `_verifyProof` into `LineaRollupBase.sol`
contracts/src/rollup/ZkEvmV2.sol
// added `constructor` and `_computeChainConfigurationHash`
contracts/src/verifiers/PlonkVerifierForDataAggregation.sol

2) changes merged from deleted branch `contract-folder-restructure`

// refactor to inherit from new file `TokenBridgeBase.sol`
// removal of `setRemoteTokenBridge`, field added to initialization data
// and `__TokenBridge_init` now calls `_setRemoteSender`
```

```
contracts/src/bridging/token/TokenBridge.sol
contracts/src/bridging/token/TokenBridgeBase.sol
contracts/src/security/reentrancy/TransientStorageReentrancyGuardUpgradeable.sol

// remove usage of `TransientStorageHelpers` instead using
// new `transient` keyword variable
contracts/src/messaging/l1/v1/L1MessageManagerV1.sol
contracts/src/messaging/l1/L1MessageService.sol

// use `TransientStorageReentrancyGuardUpgradeable` instead of `ReentrancyGuardUpgradeable`
// and use `TRANSIENT_MESSAGE_SENDER` plus deprecate `_messageSender`
contracts/src/messaging/l2/v1/L2MessageServiceV1.sol

// refactor to inherit from new file `L2MessageServiceBase.sol`
contracts/src/messaging/l2/L2MessageService.sol
contracts/src/messaging/l2/L2MessageServiceBase.sol

// security council pausing see new `SECURITY_COUNCIL_ROLE`, `PAUSE_DURATION`,
// `COOLDOWN_DURATION`, `pauseExpiryTimestamp`, `unPauseByExpiredType`
// deprecated `pauseTypeStatuses_DEPRECATED`
contracts/src/security/pausing/PauseManager.sol

3) data modularization changes related to https://github.com/Consensys/linea-monorepo/pull/1743/files

// refactored from L1MessageServiceBase
contracts/src/messaging/l1/v1/ClaimMessageV1.sol

// now inherits from LivenessRecovery, Eip4844BlobAcceptor, ClaimMessageV1
// changes/additions initialize, reinitializeV8, renounceRole
contracts/src/rollup/LineaRollup.sol

// new PROXY_ADMIN_SLOT, remove SHNARF_EXISTS_DEFAULT_VALUE, SIX_MONTHS_IN_SECONDS
// blobShnarfExists -> _blobShnarfExists made internal
// new livenessRecoveryOperator, shnarfProvider
// changes to __LineaRollup_init, moved bunch of stuff to other files
contracts/src/rollup/LineaRollupBase.sol

// new files mostly containing stuff from move LineaRollupBase with slight tweaks
contracts/src/rollup/LivenessRecovery.sol
contracts/src/rollup/Validium.sol
contracts/src/rollup/dataAvailability/CalldataBlobAcceptor.sol
contracts/src/rollup/dataAvailability/Eip4844BlobAcceptor.sol
contracts/src/rollup/dataAvailability/LocalShnarfProvider.sol
contracts/src/rollup/dataAvailability/ShnarfDataAcceptor.sol
contracts/src/rollup/dataAvailability/ShnarfDataAcceptorBase.sol
```

# 6 Executive Summary

Over the course of 5 days, the Cyfrin team conducted an audit on the Linea Mixed Upgrade smart contracts provided by Linea. In this period, a total of 8 issues were found.

The findings consist of 1 Critical and 1 Low severity issue with the remainder being informational.

### TokenBridge Storage Upgrade

Using `forge inspect -R "@openzeppelin/=contracts/node_modules/@openzeppelin/" --hardhat TokenBridge storageLayout` on both the new and old contracts to carefully examine their exact storage layout, accounting for the fix in C-1:

```
 NEW TokenBridge Layout           | Type                                    | Slot | Offset | Bytes
----------------------------------|-----------------------------------------|------|--------|------
```

```
# identical
 _initialized                     | uint8                                  | 0    | 0      | 1
 _initializing                    | bool                                   | 0    | 1      | 1

# overwrites `_status`,
# upgrade will wipe slot 1 clean
 __gap_ReentrancyGuardUpgradeable | uint256[50]                            | 1    | 0      | 1600

# identical
 __gap                            | uint256[50]                            | 51   | 0      | 1600
 __gap                            | uint256[50]                            | 101  | 0      | 1600
 _roles                           | mapping(bytes32 => struct ACU.RoleData) | 151  | 0      | 32
 __gap                            | uint256[49]                            | 152  | 0      | 1568
 messageService                   | contract IMessageService               | 201  | 0      | 20
 remoteSender                     | address                                | 202  | 0      | 20
 __base_gap                       | uint256[10]                            | 203  | 0      | 320
 pauseTypeStatuses_DEPRECATED     | mapping(bytes32 => bool)               | 213  | 0      | 32
 _pauseTypeStatusesBitMap         | uint256                                | 214  | 0      | 32
 _pauseTypeRoles                  | mapping(enum IPM.PauseType => bytes32) | 215  | 0      | 32
 _unPauseTypeRoles                | mapping(enum IPM.PauseType => bytes32) | 216  | 0      | 32
 pauseExpiryTimestamp             | uint256                                | 217  | 0      | 32
 __gap                            | uint256[6]                             | 218  | 0      | 192
 __gap_39                         | uint256[39]                            | 224  | 0      | 1248
 tokenBeacon                      | address                                | 263  | 0      | 20
 nativeToBridgedToken             | m(uint256 => m(address => address))    | 264  | 0      | 32
 bridgedToNativeToken             | mapping(address => address)            | 265  | 0      | 32
 sourceChainId                    | uint256                                | 266  | 0      | 32
 targetChainId                    | uint256                                | 267  | 0      | 32
 __gap                            | uint256[50]                            | 268  | 0      | 1600
```

```
 OLD TokenBridge Layout           | Type                                   | Slot | Offset | Bytes
----------------------------------|----------------------------------------|------|--------|------
# identical
 _initialized                     | uint8                                  | 0    | 0      | 1
 _initializing                    | bool                                   | 0    | 1      | 1

# repurposed as gap, wiped
 _status                          | uint256                                | 1    | 0      | 32

# identical
 __gap                            | uint256[50]                            | 51   | 0      | 1600
 __gap                            | uint256[50]                            | 101  | 0      | 1600
 _roles                           | mapping(bytes32 => struct ACU.RoleData) | 151  | 0      | 32
 __gap                            | uint256[49]                            | 152  | 0      | 1568
 messageService                   | contract IMessageService               | 201  | 0      | 20
 remoteSender                     | address                                | 202  | 0      | 20
 __base_gap                       | uint256[10]                            | 203  | 0      | 320
 pauseTypeStatuses_DEPRECATED     | mapping(bytes32 => bool)               | 213  | 0      | 32
 _pauseTypeStatusesBitMap         | uint256                                | 214  | 0      | 32
 _pauseTypeRoles                  | mapping(enum IPM.PauseType => bytes32) | 215  | 0      | 32
 _unPauseTypeRoles                | mapping(enum IPM.PauseType => bytes32) | 216  | 0      | 32
 pauseExpiryTimestamp             | uint256                                | 217  | 0      | 32
 __gap                            | uint256[6]                             | 218  | 0      | 192
 __gap_39                         | uint256[39]                            | 224  | 0      | 1248
 tokenBeacon                      | address                                | 263  | 0      | 20
 nativeToBridgedToken             | m(uint256 => m(address => address))    | 264  | 0      | 32
 bridgedToNativeToken             | mapping(address => address)            | 265  | 0      | 32
 sourceChainId                    | uint256                                | 266  | 0      | 32
 targetChainId                    | uint256                                | 267  | 0      | 32
 __gap                            | uint256[50]                            | 268  | 0      | 1600
```

### LineaRollup Storage Upgrade

Using `forge inspect -R "@openzeppelin/=contracts/node_modules/@openzeppelin/" --hardhat --evm-version cancun LineaRollup storageLayout` on both the new and old contracts to carefully examine their exact storage layout:

```
 NEW LineaRollup Layout                    | Type                              | Slot | Offset | Bytes
-------------------------------------------|-----------------------------------|------|--------|------
# identical
 _initialized                              | uint8                             | 0    | 0      | 1
 _initializing                             | bool                              | 0    | 1      | 1
 __gap                                     | uint256[50]                       | 1    | 0      | 1600
 __gap                                     | uint256[50]                       | 51   | 0      | 1600
 _roles                                    | m(bytes32 => struct ACU.RoleData) | 101  | 0      | 32
 __gap                                     | uint256[49]                       | 102  | 0      | 1568
 periodInSeconds                           | uint256                           | 151  | 0      | 32
 limitInWei                                | uint256                           | 152  | 0      | 32
 currentPeriodEnd                          | uint256                           | 153  | 0      | 32
 currentPeriodAmountInWei                  | uint256                           | 154  | 0      | 32
 __gap                                     | uint256[10]                       | 155  | 0      | 320
 outboxL1L2MessageStatus                   | m(bytes32 => uint256)             | 165  | 0      | 32
 inboxL2L1MessageStatus                    | m(bytes32 => uint256)             | 166  | 0      | 32
 __gap_ReentrancyGuardUpgradeable          | uint256[50]                       | 167  | 0      | 1600
 pauseTypeStatuses_DEPRECATED              | m(bytes32 => bool)                | 217  | 0      | 32
 _pauseTypeStatusesBitMap                  | uint256                           | 218  | 0      | 32
 _pauseTypeRoles                           | m(enum IPM.PauseType => bytes32)  | 219  | 0      | 32
 _unPauseTypeRoles                         | m(enum IPM.PauseType => bytes32)  | 220  | 0      | 32
 pauseExpiryTimestamp                      | uint256                           | 221  | 0      | 32
 __gap                                     | uint256[6]                        | 222  | 0      | 192
 nextMessageNumber                         | uint256                           | 228  | 0      | 32
 _messageSender_DEPRECATED                 | address                           | 229  | 0      | 20
 __gap                                     | uint256[50]                       | 230  | 0      | 1600
 currentTimestamp_DEPRECATED               | uint256                           | 280  | 0      | 32
 currentL2BlockNumber                      | uint256                           | 281  | 0      | 32
 stateRootHashes                           | mapping(uint256 => bytes32)       | 282  | 0      | 32
 verifiers                                 | mapping(uint256 => address)       | 283  | 0      | 32
 __gap                                     | uint256[50]                       | 284  | 0      | 1600
 rollingHashes                             | mapping(uint256 => bytes32)       | 334  | 0      | 32
 _messageClaimedBitMap                     | struct BitMaps.BitMap             | 335  | 0      | 32
 l2MerkleRootsDepths                       | mapping(bytes32 => uint256)       | 336  | 0      | 32
 __gap_L1MessageManager                    | uint256[50]                       | 337  | 0      | 1600
 systemMigrationBlock                      | uint256                           | 387  | 0      | 32
 __gap_L1MessageService                    | uint256[50]                       | 388  | 0      | 1600
 dataFinalStateRootHashes_DEPRECATED       | mapping(bytes32 => bytes32)       | 438  | 0      | 32
 dataParents_DEPRECATED                    | mapping(bytes32 => bytes32)       | 439  | 0      | 32
 dataShnarfHashes_DEPRECATED               | mapping(bytes32 => bytes32)       | 440  | 0      | 32
 dataStartingBlock_DEPRECATED              | mapping(bytes32 => uint256)       | 441  | 0      | 32
 dataEndingBlock_DEPRECATED                | mapping(bytes32 => uint256)       | 442  | 0      | 32
 currentL2StoredL1MessageNumber_DEPRECATED | uint256                           | 443  | 0      | 32
 currentL2StoredL1RollingHash_DEPRECATED   | bytes32                           | 444  | 0      | 32
 currentFinalizedShnarf                    | bytes32                           | 445  | 0      | 32
 _blobShnarfExists                         | mapping(bytes32 => uint256)       | 446  | 0      | 32
 currentFinalizedState                     | bytes32                           | 447  | 0      | 32

# renamed from `fallbackOperator`
 livenessRecoveryOperator                  | address                           | 448  | 0      | 20

# new, previously gap
 shnarfProvider                            | contract IProvideShnarf           | 449  | 0      | 20
 __gap_LineaRollup                         | uint256[50]                       | 450  | 0      | 1600
 __gap_LivenessRecoveryOperator            | uint256[50]                       | 500  | 0      | 1600
```

```
 OLD LineaRollup Layout                    | Type                              | Slot | Offset | Bytes
-------------------------------------------|-----------------------------------|------|--------|------
# identical
```

```
 _initialized                              | uint8                              | 0    | 0      | 1
 _initializing                             | bool                               | 0    | 1      | 1
 __gap                                     | uint256[50]                        | 1    | 0      | 1600
 __gap                                     | uint256[50]                        | 51   | 0      | 1600
 _roles                                    | m(bytes32 => struct ACU.RoleData)  | 101  | 0      | 32
 __gap                                     | uint256[49]                        | 102  | 0      | 1568
 periodInSeconds                           | uint256                            | 151  | 0      | 32
 limitInWei                                | uint256                            | 152  | 0      | 32
 currentPeriodEnd                          | uint256                            | 153  | 0      | 32
 currentPeriodAmountInWei                  | uint256                            | 154  | 0      | 32
 __gap                                     | uint256[10]                        | 155  | 0      | 320
 outboxL1L2MessageStatus                   | m(bytes32 => uint256)              | 165  | 0      | 32
 inboxL2L1MessageStatus                    | m(bytes32 => uint256)              | 166  | 0      | 32
 __gap_ReentrancyGuardUpgradeable          | uint256[50]                        | 167  | 0      | 1600
 pauseTypeStatuses_DEPRECATED              | m(bytes32 => bool)                 | 217  | 0      | 32
 _pauseTypeStatusesBitMap                  | uint256                            | 218  | 0      | 32
 _pauseTypeRoles                           | m(enum IPM.PauseType => bytes32)   | 219  | 0      | 32
 _unPauseTypeRoles                         | m(enum IPM.PauseType => bytes32)   | 220  | 0      | 32
 pauseExpiryTimestamp                      | uint256                            | 221  | 0      | 32
 __gap                                     | uint256[6]                         | 222  | 0      | 192
 nextMessageNumber                         | uint256                            | 228  | 0      | 32
 _messageSender_DEPRECATED                 | address                            | 229  | 0      | 20
 __gap                                     | uint256[50]                        | 230  | 0      | 1600
 currentTimestamp_DEPRECATED               | uint256                            | 280  | 0      | 32
 currentL2BlockNumber                      | uint256                            | 281  | 0      | 32
 stateRootHashes                           | mapping(uint256 => bytes32)        | 282  | 0      | 32
 verifiers                                 | mapping(uint256 => address)        | 283  | 0      | 32
 __gap                                     | uint256[50]                        | 284  | 0      | 1600
 rollingHashes                             | mapping(uint256 => bytes32)        | 334  | 0      | 32
 _messageClaimedBitMap                     | struct BitMaps.BitMap              | 335  | 0      | 32
 l2MerkleRootsDepths                       | mapping(bytes32 => uint256)        | 336  | 0      | 32
 __gap_L1MessageManager                    | uint256[50]                        | 337  | 0      | 1600
 systemMigrationBlock                      | uint256                            | 387  | 0      | 32
 __gap_L1MessageService                    | uint256[50]                        | 388  | 0      | 1600
 dataFinalStateRootHashes_DEPRECATED       | mapping(bytes32 => bytes32)        | 438  | 0      | 32
 dataParents_DEPRECATED                    | mapping(bytes32 => bytes32)        | 439  | 0      | 32
 dataShnarfHashes_DEPRECATED               | mapping(bytes32 => bytes32)        | 440  | 0      | 32
 dataStartingBlock_DEPRECATED              | mapping(bytes32 => uint256)        | 441  | 0      | 32
 dataEndingBlock_DEPRECATED                | mapping(bytes32 => uint256)        | 442  | 0      | 32
 currentL2StoredL1MessageNumber_DEPRECATED | uint256                            | 443  | 0      | 32
 currentL2StoredL1RollingHash_DEPRECATED   | bytes32                            | 444  | 0      | 32
 currentFinalizedShnarf                    | bytes32                            | 445  | 0      | 32
 _blobShnarfExists                         | mapping(bytes32 => uint256)        | 446  | 0      | 32
 currentFinalizedState                     | bytes32                            | 447  | 0      | 32

# renamed to `livenessRecoveryOperator`
 fallbackOperator                          | address                            | 448  | 0      | 20

# 449 - used for new `shnarfProvider`
 __gap_LineaRollup                         | uint256[50]                        | 449  | 0      | 1600
```

### L2MessageService Storage Upgrade

Using `forge inspect -R "@openzeppelin/=contracts/node_modules/@openzeppelin/" --hardhat L2MessageService storageLayout` on both the new and old contracts to carefully examine their exact storage layout:

```
 NEW L2MessageService Layout      | Type                                   | Slot | Offset | Bytes
----------------------------------|----------------------------------------|------|--------|------
# identical
 _initialized                     | uint8                                  | 0    | 0      | 1
 _initializing                    | bool                                   | 0    | 1      | 1
 __gap                            | uint256[50]                            | 1    | 0      | 1600
```

```
 __gap                            | uint256[50]                              | 51   | 0      | 1600
 _roles                           | mapping(bytes32 => struct ACU.RoleData) | 101  | 0      | 32
 __gap                            | uint256[49]                              | 102  | 0      | 1568
 periodInSeconds                  | uint256                                  | 151  | 0      | 32
 limitInWei                       | uint256                                  | 152  | 0      | 32
 currentPeriodEnd                 | uint256                                  | 153  | 0      | 32
 currentPeriodAmountInWei         | uint256                                  | 154  | 0      | 32
 __gap                            | uint256[10]                              | 155  | 0      | 320
 pauseTypeStatuses_DEPRECATED     | mapping(bytes32 => bool)                 | 165  | 0      | 32
 _pauseTypeStatusesBitMap         | uint256                                  | 166  | 0      | 32
 _pauseTypeRoles                  | mapping(enum IPM.PauseType => bytes32)   | 167  | 0      | 32
 _unPauseTypeRoles                | mapping(enum IPM.PauseType => bytes32)   | 168  | 0      | 32
 pauseExpiryTimestamp             | uint256                                  | 169  | 0      | 32
 __gap                            | uint256[6]                               | 170  | 0      | 192
 inboxL1L2MessageStatus           | mapping(bytes32 => uint256)              | 176  | 0      | 32

# overwrites `_status`,
# upgrade will wipe slot 177 clean
 __gap_ReentrancyGuardUpgradeable | uint256[50]                              | 177  | 0      | 1600

# identical
 __gap_L2MessageServiceV1         | uint256[50]                              | 227  | 0      | 1600

# renamed
 _messageSender_DEPRECATED        | address                                  | 277  | 0      | 20

# identical
 nextMessageNumber                | uint256                                  | 278  | 0      | 32
 minimumFeeInWei                  | uint256                                  | 279  | 0      | 32
 lastAnchoredL1MessageNumber      | uint256                                  | 280  | 0      | 32
 l1RollingHashes                  | mapping(uint256 => bytes32)              | 281  | 0      | 32
 __gap_L2MessageManager           | uint256[50]                              | 282  | 0      | 1600
 __gap_L2MessageService           | uint256[50]                              | 332  | 0      | 1600
 __gap_L2MessageService           | uint256[50]                              | 382  | 0      | 1600
```

```
 OLD L2MessageService Layout      | Type                                     | Slot | Offset | Bytes
----------------------------------|------------------------------------------|------|--------|------
# identical
 _initialized                     | uint8                                    | 0    | 0      | 1
 _initializing                    | bool                                     | 0    | 1      | 1
 __gap                            | uint256[50]                              | 1    | 0      | 1600
 __gap                            | uint256[50]                              | 51   | 0      | 1600
 _roles                           | mapping(bytes32 => struct ACU.RoleData) | 101  | 0      | 32
 __gap                            | uint256[49]                              | 102  | 0      | 1568
 periodInSeconds                  | uint256                                  | 151  | 0      | 32
 limitInWei                       | uint256                                  | 152  | 0      | 32
 currentPeriodEnd                 | uint256                                  | 153  | 0      | 32
 currentPeriodAmountInWei         | uint256                                  | 154  | 0      | 32
 __gap                            | uint256[10]                              | 155  | 0      | 320
 pauseTypeStatuses_DEPRECATED     | mapping(bytes32 => bool)                 | 165  | 0      | 32
 _pauseTypeStatusesBitMap         | uint256                                  | 166  | 0      | 32
 _pauseTypeRoles                  | mapping(enum IPM.PauseType => bytes32)   | 167  | 0      | 32
 _unPauseTypeRoles                | mapping(enum IPM.PauseType => bytes32)   | 168  | 0      | 32
 pauseExpiryTimestamp             | uint256                                  | 169  | 0      | 32
 __gap                            | uint256[6]                               | 170  | 0      | 192
 inboxL1L2MessageStatus           | mapping(bytes32 => uint256)              | 176  | 0      | 32

# overwritten by `__gap_ReentrancyGuardUpgradeable`
 _status                          | uint256                                  | 177  | 0      | 32
 __gap                            | uint256[49]                              | 178  | 0      | 1568

# identical
 __gap_L2MessageServiceV1         | uint256[50]                              | 227  | 0      | 1600
```

```
# renamed to `_messageSender_DEPRECATED`
 _messageSender                  | address                                | 277  | 0      | 20

# identical
 nextMessageNumber               | uint256                                | 278  | 0      | 32
 minimumFeeInWei                 | uint256                                | 279  | 0      | 32
 lastAnchoredL1MessageNumber     | uint256                                | 280  | 0      | 32
 l1RollingHashes                 | mapping(uint256 => bytes32)            | 281  | 0      | 32
 __gap_L2MessageManager          | uint256[50]                            | 282  | 0      | 1600
 __gap_L2MessageService          | uint256[50]                            | 332  | 0      | 1600
 __gap_L2MessageService          | uint256[50]                            | 382  | 0      | 1600
```

### Merged To Main

The new functionality containing mitigations from our audit was merged to `main` branch in PR2007 and commit 5074237.

### Deployment Bytecode Verification

After an additional audit by another provider, the resulting code was subsequently deployed on-chain to:

- LineaRollup: 0xE68697690E8ff196A6aBB3E1385156D87Df85332
- L1 TokenBridge: 0xF0e003F0dE2d583Ae28FA8cBF66aa096CdAce3ff
- L2MessageService: 0x9976fD7edDb78156a002DE74c9158E884702273d
- L2 TokenBridge: 0x4a496167F187A97379e763f693A499cE1182848b

At commit 698d8830c, Cyfrin verified that the locally compiled bytecode matched both the expected bytecode in the repository and the deployed bytecode on-chain. We used the following reproducible process to do this.

1) prepare the repo:

```
gh repo clone Consensys/linea-monorepo
cd linea-monorepo
git checkout 698d8830c359b698ceee510f0d2beedebac9f7fe
pnpm install --ignore-scripts // to get around an error related to better-sqlite3
cd node_modules/.pnpm/c-kzg@4.1.0/node_modules/c-kzg && npx node-gyp rebuild // rebuild c-kzg native
↪  bindings
cd contracts
npx hardhat test
```

2) diff the locally compiled bytecode against the expected:

```
linea-monorepo/contracts$ diff build/src/rollup/LineaRollup.sol/LineaRollup.json
↪  deployments/bytecode/2026-02-17/LineaRollup.json

linea-monorepo/contracts$ diff build/src/messaging/l2/L2MessageService.sol/L2MessageService.json
↪  deployments/bytecode/2026-02-17/L2MessageService.json

linea-monorepo/contracts$ diff build/src/bridging/token/TokenBridge.sol/TokenBridge.json
↪  deployments/bytecode/2026-02-17/TokenBridge.json
```

3) prepare the helper programs:

To extract deployment bytecode use this python program `extract_deployed_bytecode.py`:

```python
import sys
import os
import requests

if len(sys.argv) != 4:
    print("Usage: python3 extract_deployed_bytecode.py <env_var_for_rpc_url> <address> <output_path>")
    sys.exit(1)
```

```
env_var = sys.argv[1]
address = sys.argv[2]
output_path = sys.argv[3]

rpc_url = os.environ.get(env_var)
if not rpc_url:
    print(f"Error: Environment variable '{env_var}' is not set.")
    sys.exit(1)

data = {
    "jsonrpc": "2.0",
    "method": "eth_getCode",
    "params": [address, "latest"],
    "id": 1
}

response = requests.post(rpc_url, json=data)
if response.status_code == 200:
    result = response.json()
    if "result" in result:
        bytecode = result["result"]
        with open(output_path, "w") as f:
            f.write(bytecode)
        print(f"Deployed bytecode saved to {output_path}")
    else:
        print("Error: No bytecode found in response")
else:
    print(f"Error: RPC request failed with status {response.status_code}")
```

To compare locally compiled bytecode against deployed bytecode ignoring differences in metadata & immutable positions, use this python program `compare_bytecodes.py`:

```
import sys
import json
import os

def strip_metadata(bytecode: bytes) -> bytes:
    if len(bytecode) < 2:
        return bytecode
    cbor_len = (bytecode[-2] << 8) | bytecode[-1]
    if cbor_len <= 0 or cbor_len + 2 > len(bytecode):
        return bytecode
    potential_cbor = bytecode[-cbor_len - 2 : -2]
    if potential_cbor[0] not in (0xa1, 0xa2):
        return bytecode
    return bytecode[:-cbor_len - 2]

if len(sys.argv) != 3:
    print("Usage: python3 compare_bytecodes.py <artifact_json_path> <deployed_bytecode_path>")
    sys.exit(1)

artifact_path = sys.argv[1]
deployed_path = sys.argv[2]

try:
    with open(artifact_path, "r") as f:
        data = json.load(f)

    deployed_bytecode_section = data.get("deployedBytecode")
    if isinstance(deployed_bytecode_section, str):
        local_hex = deployed_bytecode_section
    elif isinstance(deployed_bytecode_section, dict):
```

```
        local_hex = deployed_bytecode_section.get("object")
    else:
        print("Error: 'deployedBytecode' not found or invalid in artifact.")
        sys.exit(1)

    if not local_hex:
        print("Error: No bytecode found in artifact.")
        sys.exit(1)

    if local_hex.startswith("0x"):
        local_hex = local_hex[2:]

    local_bytes = bytes.fromhex(local_hex)

    with open(deployed_path, "r") as f:
        deployed_hex = f.read().strip()

    if deployed_hex.startswith("0x"):
        deployed_hex = deployed_hex[2:]

    deployed_bytes = bytes.fromhex(deployed_hex)

    local_bytes = strip_metadata(local_bytes)
    deployed_bytes = strip_metadata(deployed_bytes)

    if len(local_bytes) != len(deployed_bytes):
        print(f"Error: Bytecode lengths differ after stripping metadata: local {len(local_bytes)} vs
        ↪  deployed {len(deployed_bytes)}")
        sys.exit(1)

    mismatch = False
    assumed_immutable_positions = []
    for i in range(len(local_bytes)):
        if local_bytes[i] == 0 and deployed_bytes[i] != 0:
            assumed_immutable_positions.append(i)
            continue
        if local_bytes[i] != deployed_bytes[i]:
            print(f"Mismatch at position {i}: local 0x{local_bytes[i]:02x} vs deployed
            ↪  0x{deployed_bytes[i]:02x}")
            mismatch = True

    if mismatch:
        print("Bytecodes do NOT match (differences outside assumed immutable positions).")
    else:
        print("Bytecodes match, ignoring differences in assumed immutable positions.")

    if assumed_immutable_positions:
        print("\nAssumed immutable positions where local=0x00 and deployed !=0x00:")
        groups = []
        start = assumed_immutable_positions[0]
        prev = assumed_immutable_positions[0]
        for pos in assumed_immutable_positions[1:]:
            if pos == prev + 1:
                prev = pos
            else:
                groups.append((start, prev))
                start = pos
                prev = pos
        groups.append((start, prev))
        for s, e in groups:
            length = e - s + 1
            local_val = local_bytes[s:s+length].hex()
            deployed_val = deployed_bytes[s:s+length].hex()
```

```
            print(f"  Range {s}-{e} (length {length}): local {local_val} vs deployed {deployed_val}")

except FileNotFoundError as e:
    print(f"Error: File not found - {e}")
    sys.exit(1)
except json.JSONDecodeError:
    print(f"Error: Invalid JSON in artifact file '{artifact_path}'.")
    sys.exit(1)
except ValueError as e:
    print(f"Error: Invalid hex in bytecode - {e}")
    sys.exit(1)
except Exception as e:
    print(f"Unexpected error: {str(e)}")
    sys.exit(1)
```

4) verify `LineaRollup` deployed bytecode using:

- `python3 extract_deployed_bytecode.py ETH_RPC_URL 0xE68697690E8ff196A6aBB3E1385156D87Df85332 deployments/bytecode/2026-02-17/LineaRollup.mainnet.txt` to extract mainnet deployed bytecode
- `python3 compare_bytecodes.py build/src/rollup/LineaRollup.sol/LineaRollup.json deployments/bytecode/2026-02-17/LineaRollup.mainnet.txt` to verify both match - succeeds

5) verify `L1 TokenBridge` deployed bytecode using:

- `python3 extract_deployed_bytecode.py ETH_RPC_URL 0xF0e003F0dE2d583Ae28FA8cBF66aa096CdAce3ff deployments/bytecode/2026-02-17/TokenBridge.L1.mainnet.txt` to extract mainnet deployed bytecode
- `python3 compare_bytecodes.py build/src/bridging/token/TokenBridge.sol/TokenBridge.json deployments/bytecode/2026-02-17/TokenBridge.L1.mainnet.txt` to verify both match - succeeds

6) verify `L2MessageService` deployed bytecode using:

- `python3 extract_deployed_bytecode.py LINEA_RPC_URL 0x9976fD7edDb78156a002DE74c9158E884702273d deployments/bytecode/2026-02-17/L2MessageService.mainnet.txt` to extract Linea mainnet deployed bytecode (public RPC: `https://rpc.linea.build`)
- `python3 compare_bytecodes.py build/src/messaging/l2/L2MessageService.sol/L2MessageService.json deployments/bytecode/2026-02-17/L2MessageService.mainnet.txt` to verify both match - succeeds

7) verify `L2 TokenBridge` deployed bytecode using:

- `python3 extract_deployed_bytecode.py LINEA_RPC_URL 0x4a496167F187A97379e763f693A499cE1182848b deployments/bytecode/2026-02-17/TokenBridge.L2.mainnet.txt` to extract Linea mainnet deployed bytecode (public RPC: `https://rpc.linea.build`)
- `python3 compare_bytecodes.py build/src/bridging/token/TokenBridge.sol/TokenBridge.json deployments/bytecode/2026-02-17/TokenBridge.L2.mainnet.txt` to verify both match - succeeds

## Summary

| | |
|---|---|
| Project Name | Linea Mixed Upgrade |
| Repository | linea-monorepo |
| Commit | 78f9d1fa3ccc... |
| Fix Commit | c374a2c9f3a6... |
| Audit Timeline | Jan 5th - Jan 9th, 2026 |
| Methods | Manual Review |

**Issues Found**

| | |
|---|---|
| Critical Risk | 1 |
| High Risk | 0 |
| Medium Risk | 0 |
| Low Risk | 1 |
| Informational | 6 |
| Gas Optimizations | 0 |
| Total Issues | 8 |

**Summary of Findings**

| | |
|---|---|
| [C-1] After the upgrade permissionless attacker can fully drain the L1 `TokenBridge` of `ERC20` tokens currently valued around $29MUSD$ | Resolved |
| [L-1] `SECURITY_COUNCIL_ROLE` unpausing a type leads to automatically marking as expired any other pause types, whether they were enacted by the `SECURITY_COUNCIL_ROLE` or not | Resolved |
| [I-1] `LineaRollup, LivenessRecovery::renounceRole` prevents liveness recovery operator from renouncing all roles | Acknowledged |
| [I-2] `LivenessRecovery::setLivenessRecoveryOperator` will emit misleading event when role is not granted | Resolved |
| [I-3] Inconsistent handling of update/set transactions which don't actually change values | Acknowledged |
| [I-4] Not emitting event to log the version's change when reinitializing the `LineaRollup` contract | Resolved |
| [I-5] Missing `onlyInitializing` modifier on initialization functions for abstract contracts | Resolved |
| [I-6] Consider wiping slot 177 on Linea `L2MessageService` after upgrade | Resolved |

# 7 Findings

## 7.1 Critical Risk

### 7.1.1 After the upgrade permissionless attacker can fully drain the L1 `TokenBridge` of `ERC20` tokens currently valued around $29M USD

**Description:** Using `forge inspect -R "@openzeppelin/=contracts/node_modules/@openzeppelin/" --hardhat TokenBridge storageLayout` on both the new and old `TokenBridge` contracts to carefully examine their exact storage layout showed that:

- slot 0 which used to be initialization slot becomes a gap
- slot 50 which used to be a gap becomes the new initialization slot

**Impact:** Immediately following the upgrade, the `TokenBridge` contract will believe it is not initialized allowing a permissionless attacker to initialize it. An attacker can weaponize this to completely drain the L1 `TokenBridge` contract of ERC20 tokens which at the time of this audit are valued around $29M USD.

**Proof of Concept:** Immediately following the upgrade:

1. Attacker calls `TokenBridge::initialize` to set themselves as default admin
2. Attacker calls `TokenBridge::grantRole(SET_MESSAGE_SERVICE_ROLE, attacker)` to give themselves permission to set messaging service address
3. Attacker deploys a malicious messaging service contract such as:

```
contract MaliciousMessageService {
    address public targetRemoteSender;

    constructor(address _remoteSender) {
        targetRemoteSender = _remoteSender;
    }

    // Spoofs the remoteSender check
    function sender() external view returns (address) {
        return targetRemoteSender;
    }

    // Calls TokenBridge as msg.sender to pass onlyMessagingService
    function drain(
        ITokenBridge bridge,
        address token,
        uint256 amount,
        address recipient,
        uint256 chainId
    ) external {
        bridge.completeBridging(token, amount, recipient, chainId, "");
    }
}
```

4. Attacker sets it by calling `TokenBridge::setMessageService(maliciousMessageService)`
5. Attacker calls the `drain` function on their malicious messaging service for every token locked in the L1 `TokenBridge`
6. `TokenBridge::_completeBridging` calls `IERC20Upgradeable(_nativeToken).safeTransfer(_recipient, _amount)` to send the attacker the locked tokens

This attack can be executed atomically and via a private mempool such as flashbots (to prevent front-running) making it unstoppable and completely draining the L1 `TokenBridge`.

**Recommended Mitigation:** `TokenBridgeBase` should inherit from `Initializable`. In an older version `TokenBridge` inherited from `Initializable` but this was later changed to inherit from OZ `ReentrancyGuardUpgradeable`,

which itself inherits from `Initializable` so everything was still OK.

The bug appears to have been introduced on Nov 7th 2025 in commit 0c8bee7 which swapped out OZ `ReentrancyGuardUpgradeable` for the new custom `TransientStorageReentrancyGuardUpgradeable`. This new contract doesn't inherit from `Initializable` which changed the `TokenBridge` inheritance hierarchy and hence storage slots.

**Linea:** Fixed in commit 4882f33.

**Cyfrin:** Verified; the fix results in the initialization slot being preserved at slot 0. Slot 1 which used to be `_status` now becomes a gap and using `cast storage 0x051F1D88f0aF5763fB888eC4378b4D8B29ea3319 1` shows that on Mainnet L1 `TokenBridge`, slot 1 (currently `_status`) is already set to 1. Consider using a `reinitializer` to wipe slot 1 "clean" as it becomes a gap.

**Linea:** Added the wiping of slot 1 in commit d99f590.

## 7.2 Low Risk

### 7.2.1 SECURITY_COUNCIL_ROLE unpausing a type leads to automatically marking as expired any other pause types, whether they were enacted by the SECURITY_COUNCIL_ROLE or not

**Description:** PauseManager enables the handling of the pausing/unpausing of different PauseTypes, targeting specific system functionalities (as well as a GENERAL pause). There are two main types of pausers:

1. Pausers with theSECURITY_COUNCIL_ROLE
2. Pausers without the SECURITY_COUNCIL_ROLE

The main difference between the two is that pausers with SECURITY_COUNCIL_ROLE can pause without cooldown or expiry restrictions, and when they unpause, the pauseExpiryTimestamp is reset to enable non-SECURITY_-COUNCIL_ROLE pausing.

There is an edge case that allows for immediately unpause of any active pause after the SECURITY_COUNCIL_ROLE unpauses one type. Whether the SECURITY_COUNCIL_ROLE unpauses a pause enacted by themselves or by a non-SECURITY_COUNCIL_ROLE, the result is the same; any other active pause can be immediately unpaused.

**Impact:** When the SECURITY_COUNCIL_ROLE unpauses a type, all the other active pause types will be immediately marked as expired, regardless of whether they were enacted by the SECURITY_COUNCIL_ROLE or a non-SECURITY_-COUNCIL_ROLE account.

**Proof of Concept:** Add the next PoC to PauseManager.ts:

```
it.only("Non-SECURITY_COUNCIL_ROLE pause L1_L2_PAUSE_TYPE -> SECURITY_COUNCIL_ROLE pause
↪  GENERAL_PAUSE_TYPE -> SECURITY_COUNCIL_ROLE unpause L1_L2_PAUSE_TYPE => GENERAL_PAUSE can be
↪  unpaused even though it had not been actually unpaused", async () => {
  await pauseByType(L1_L2_PAUSE_TYPE);
  await pauseByType(GENERAL_PAUSE_TYPE, securityCouncil);
  await unPauseByType(L1_L2_PAUSE_TYPE, securityCouncil);
  //@audit-info => GENERAL_PAUSE can be immediately unpaused even though it had not been actually
  ↪  unpaused
  await unPauseByExpiredType(GENERAL_PAUSE_TYPE, nonManager);
  expect(await pauseManager.isPaused(GENERAL_PAUSE_TYPE)).to.be.false;
  expect(await pauseManager.isPaused(L1_L2_PAUSE_TYPE)).to.be.false;
});
```

**Recommended Mitigation:** Consider not resetting the pauseExpiryTimestamp below the block.timestamp, potentially add 1 hour cooldown period from the current block.timestamp, this will prevent immediately marking other pauses as expired.

Alternatively, consider adding a bool flag to unpauseByType flag that can allow the SECURITY_COUNCIL_ROLE to select dynamically whether they want to reset the pauseExpiryTimestamp or not.

- A more elaborate alternative would be to track the active pauses and reset the pauseExpiryTimestamp only when the last active pause is unpaused by the SECURITY_COUNCIL_ROLE.

**Linea:** Fixed in PR 2335.

**Cyfrin:** Verified. Pause expirations are now tracked per pause type. Non-SecurityCouncil can't pause a pause type already paused by the SecurityCouncil, but the SecurityCouncil can pause a pause type already paused by a non-SecurityCouncil. Pauses enacted by the SecurityCouncil can only be unpaused by them. Unpausing a type only resets the expiry timestamp for that specific type.

## 7.3 Informational

### 7.3.1 `LineaRollup`, `LivenessRecovery::renounceRole` prevents liveness recovery operator from renouncing all roles

**Description:** The intention appears to be that the liveness recovery operator shouldn't be able to renounce the `OPERATOR_ROLE` they are granted, however `LivenessRecovery::renounceRole` called by `LineaRollup::renounceRole` prevents the liveness recovery operator from renouncing *all* roles:

```
function renounceRole(bytes32 _role, address _account) public virtual override {
  // @audit only checks address, not role being renounced
  if (_account == livenessRecoveryOperator) {
    revert OnlyNonLivenessRecoveryOperator();
  }

  super.renounceRole(_role, _account);
}
```

**Impact:** If the liveness recovery operator has other legitimate roles they wish to renounce, they will be unable to do so.

**Recommended Mitigation:** `LivenessRecovery::renounceRole` should only revert if `OPERATOR_ROLE` is being renounced:

```
function renounceRole(bytes32 _role, address _account) public virtual override {
- if (_account == livenessRecoveryOperator) {
+ if (_account == livenessRecoveryOperator && _role == OPERATOR_ROLE) {
    revert OnlyNonLivenessRecoveryOperator();
  }
  super.renounceRole(_role, _account);
}
```

**Linea:** Acknowledged; the liveness operator role should never and would never be granted anything other than the operator role.

### 7.3.2 `LivenessRecovery::setLivenessRecoveryOperator` will emit misleading event when role is not granted

**Description:** `LivenessRecovery::setLivenessRecoveryOperator` can be called multiple times as long as the first two preconditions are met.

However if `OPERATOR_ROLE` has already been granted to `livenessRecoveryOperator` then `AccessControlUpgradeable::_grantRole` returns `false`.

But the boolean return value of `_grantRole` is not checked so the misleading event will still be emitted.

**Recommended Mitigation:** Only emit the event if `_grantRole` returned true:

```
-   _grantRole(OPERATOR_ROLE, livenessRecoveryOperatorAddress);
+   if(_grantRole(OPERATOR_ROLE, livenessRecoveryOperatorAddress))
    emit LivenessRecoveryOperatorRoleGranted(msg.sender, livenessRecoveryOperatorAddress);
```

**Linea:** Fixed in commit 66050d2.

**Cyfrin:** Verified.

### 7.3.3 Inconsistent handling of update/set transactions which don't actually change values

**Description:** `PauseManager::updatePauseTypeRole` reverts if the previous and new roles are identical, and only writes to storage and emits an event if the value was actually changed:

```
function updatePauseTypeRole(
  PauseType _pauseType,
```

```
  bytes32 _newRole
) external onlyUsedPausedTypes(_pauseType) onlyRole(SECURITY_COUNCIL_ROLE) {
  bytes32 previousRole = _pauseTypeRoles[_pauseType];
  if (previousRole == _newRole) {
    revert RolesNotDifferent();
  }

  _pauseTypeRoles[_pauseType] = _newRole;
  emit PauseTypeRoleUpdated(_pauseType, _newRole, previousRole);
}
```

In contrast the following places don't revert on "no change" transactions, writing to storage and emitting events even if no change occurred:

- `LineaRollupBase::setVerifierAddress`

```
function setVerifierAddress(address _newVerifierAddress, uint256 _proofType) external
↪  onlyRole(VERIFIER_SETTER_ROLE) {
    if (_newVerifierAddress == address(0)) {
      revert ZeroAddressNotAllowed();
    }
    // no revert if _newVerifierAddress == verifiers[_proofType]
    emit VerifierAddressChanged(_newVerifierAddress, _proofType, msg.sender, verifiers[_proofType]);
    verifiers[_proofType] = _newVerifierAddress;
}
```

- `LineaRollupBase::unsetVerifierAddress`

```
function unsetVerifierAddress(uint256 _proofType) external onlyRole(VERIFIER_UNSETTER_ROLE) {
    // no revert if verifiers[_proofType] == address(0)
    emit VerifierAddressChanged(address(0), _proofType, msg.sender, verifiers[_proofType]);
    delete verifiers[_proofType];
}
```

- `L2MessageServiceV1::setMinimumFee`

```
function setMinimumFee(uint256 _feeInWei) external onlyRole(MINIMUM_FEE_SETTER_ROLE) {
    // no revert if _feeInWei == previousMinimumFee
    uint256 previousMinimumFee = minimumFeeInWei;
    minimumFeeInWei = _feeInWei;
    emit MinimumFeeChanged(previousMinimumFee, _feeInWei, msg.sender);
}
```

- `TokenBridgeBase::setMessageService`

```
function setMessageService(address _messageService) external ... {
    // no revert if _messageService == oldMessageService
    address oldMessageService = address(messageService);
    messageService = IMessageService(_messageService);
    emit MessageServiceUpdated(_messageService, oldMessageService, msg.sender);
}
```

- `RateLimiter::resetAmountUsedInPeriod`

```
function resetAmountUsedInPeriod() external onlyRole(USED_RATE_LIMIT_RESETTER_ROLE) {
    // no revert if currentPeriodAmountInWei == 0
    currentPeriodAmountInWei = 0;
    emit AmountUsedInPeriodReset(_msgSender());
}
```

**Recommended Mitigation:** This can be acknowledged or behavior can be harmonized if there is no specific reasons for one function to differ in behavior from others.

**Linea:** Acknowledged.

### 7.3.4 Not emitting event to log the version's change when reinitializing the `LineaRollup` contract

**Description:** When `LineaRollup` gets upgraded, `LineaRollup::reinitializeV8` is called to reinitialize permissions, roles, set the `shnarfProvider`, and bump up the `initialized` version, but `LineaRollupVersionChanged` event is not emitted to log the version's change.

**Recommended Mitigation:** Emit the event `LineaRollupVersionChanged` with the respective `previousVersion` and `newVersion` for the `LineaRollup`.

**Linea:** Fixed in PR2020.

**Cyfrin:** Verified.

### 7.3.5 Missing `onlyInitializing` modifier on initialization functions for abstract contracts

**Description:** The `onlyInitializing` modifier is the established standard for protecting internal initialization functions in abstract contracts against unintended calls post-initialization. These new initialization functions introduced here do not include this modifier, which deviates from common security patterns:

- `L2MessageServiceBase::__L2MessageService_init`
- `LineaRollupBase::__LineaRollup_init`
- `TokenBridgeBase::__TokenBridge_init`

**Recommended Mitigation:** Consider adding the `onlyInitializing` modifier to those functions.

**Linea:** Fixed in commits 802cf72, 2d63895.

**Cyfrin:** Verified.

### 7.3.6 Consider wiping slot 177 on Linea `L2MessageService` after upgrade

**Description:** After the upgrade, `L2MessageService` repurposes slot 177 for `__gap_ReentrancyGuardUpgradeable` but previously this was used for `_status`.

Using `cast storage 0x508Ca82Df566dCD1B0DE8296e70a96332cD644ec 177 --rpc-url https://rpc.linea.build` shows that slot 177 has a value of 1, so ideally this would be wiped to clean it when changing the usage of this slot into a gap.

**Linea:** Fixed in commit c462da0.

**Cyfrin:** Verified.